
ML-DSA.Verify(pk, M, σ, ctx)

Verifies a signature σ for a message M .

Input: Public key $pk \in \mathbb{B}^{32+32k(\text{bitlen}(q-1)-d)}$, message $M \in \{0, 1\}^*$, signature $\sigma \in \mathbb{B}^{\lambda/4+\ell \cdot 32 \cdot (1+\text{bitlen}(\gamma_1-1))+\omega+k}$, context string ctx (a byte string of 255 or fewer bytes).

Output: Boolean.

- 1: **if** $|ctx| > 255$ **then**
 - 2: **return** $\perp$ $\triangleright$ return an error indication if the context string is too long
 - 3: **end if**
 - 4:
 - 5: $M' \leftarrow \text{BytesToBits}(\text{IntegerToBytes}(0, 1) \parallel \text{IntegerToBytes}(|ctx|, 1) \parallel ctx) \parallel M$
 - 6: **return** ML-DSA.Verify_internal(pk, M', σ)
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